Fiscal Year:	FY 2012	**Task Last Updated:**	FY 09/29/2011
PI Name:	Cromwell, Ronita Ph.D.		
Project Title:	Bed Rest Standard Measures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	October 2010: Buccello-Stout, Regina (Universities Space Research Association) Removed from protocol. Zanello, Susana (Universities Space Research Association) Added to protocol.		
COI Name (Institution):	Ware, David (University of Texas Medical Branch) Yarbough, Patrice (Universities Space Research Association) Zanello, Susana (Universities Space Research Association)		
Grant/Contract No.:	Directed Research		
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Task Description:

The Flight Analogs Project Plan outlines a multi-phase approach for implementation of NASA-funded bed rest studies. This study is currently conducted at the NASA Flight Analogs Research Unit located at The University of Texas Medical Branch (UTMB) in Galveston, Texas. Although head-down bed rest has been used as a spaceflight analog to study changes in physiologic function associated with spaceflight for many years at many different institutions, standards for conducting such studies have not previously been established. Conditions of bed rest have been investigation or site specific. Most studies have had no direct comparisons of bed rest data with spaceflight data. As a consequence, systematic, multi-disciplinary comparisons across studies and comparisons of bed rest and actual spaceflight data have not been possible. To address these issues, the FAP has developed standard conditions for implementation of all NASA-funded bed rest studies and a set of standard measurements to be collected from all bed rest subjects. The standard measurements replicate, for the most part, measurements obtained on astronauts before and after long duration missions. Spaceflight and bed rest measurements are taken by the Johnson Space Center Laboratories and are used for three purposes: they make possible a direct comparison between responses to spaceflight and responses to head-down bed rest; they provide a characterization of human responses to long duration head-down bed rest; and they provide the ability to assess candidate countermeasures in a multi-disciplinary manner by evaluating the effects on the targeted systems as well as possible unexpected effects on non-targeted systems. The following measures are included in the standard measures complement. Nutritional status and immune function are assessed through blood, urine and saliva samples. Standard clinical laboratory assessment of blood and urine are used to monitor subjects' health status throughout the study. Dual Energy X-Ray Absorptiometry (DXA) in combination with bone biomarkers are used to assess changes in bone. Physical Fitness is assessed through measures of muscle strength, endurance, flexibility and maximum aerobic capacity. Posturography and monosynaptic T-reflex testing provides assessments of neurological function. The cardiovascular system is assessed through measures of orthostatic tolerance, blood volume and cardiac function. These multidisciplinary measures are provided to investigators to enhance interpretation of system-specific results. By standardizing conditions and measures for all bed rest studies, NASA is able to compare results across bed rest studies in a multidisciplinary manner. The standard set of measures will be collected on as many bed rest subjects as is feasible to provide a multi-discipline evaluation of potential countermeasures. These measures will be integrated with the peer-reviewed science requirements on a non-interference basis.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

In FY2011, standard measures were used in the 30-day Bed Rest Complement. Included in this report is a summary of the findings from this bed rest complement. A total of 12 healthy participants completed 30 days of 6° head-down tilt bed rest for this study. Upon admission to the NASA Flight Analogs Research Unit, participants completed 2 weeks of pre-bed rest baseline testing. This was followed by the 30-day head-down tilt bed rest period. Following the bed rest phase, participants spent 2 weeks in post bed rest testing and rehabilitation.
Investigator studies in this complement included studies entitled: Rapid Measurements of Bone Loss using Tracer-less Calcium Isotope Analysis of Blood and Urine; Validation of Near Infrared Spectroscopy (NIRS) Measures following Bed Rest; Non-invasive Device for Measuring Core Temperature during Maximal Exercise; and Methods for the Assessment of Gastrointestinal Physiology and Function in a Reduced Gravity Analog.

Standard measures were integrated with these investigator studies. These standard measures included measures from the following disciplines: cardiovascular, neurological, exercise, and nutritional assessments. A summary of results from these tests follows.

Cardiovascular Measures

Plasma Volume Index (PVI) provides an assessment of fluid shifts as the body adapts to the position of 6 degrees head-down tilt. This adaptation is similar to what astronauts experience in space. PVI values during bed rest and immediately post bed rest decreased significantly from the pre-bed rest value. PVI recovered to pre-bed rest values by the fourth day after bed rest. Shifting of body fluids produces orthostatic intolerance (light headedness or fainting upon standing) when subjects are placed in a near vertical position, as in the Cardiovascular Tilt Test. Results of this test indicated that subjects have a significantly lower probability of completing the tilt test on the day they get up from bed as compared to pre-bed rest. Subjects recover quickly as the probability of completing the tilt test on the fourth day out of bed is not different from pre-bed rest.

Neurological Measures

Sensory Organization Tests (SOTs) are used to assess postural stability before and after bed rest. The SOTs objectively assess the ability to make effective use of neuro-sensory information to maintain upright posture. The most challenging SOTs involve maintaining upright posture while making head movements. Results demonstrated that postural stability was significantly challenged on the first day after bed rest, but recovers quickly by the third day out of bed.

T-reflex testing assesses the integrity of reflexes in lower extremity muscles that control upright posture. As early as five days into the bed rest period, the nerves that supply these muscles exhibit increased reflex latency and decreased conduction velocity. In other words, the reflex is slower to respond and the speed at which the signal travels is slower. These changes continued to progress throughout bed rest and became significantly by the final day of bed rest. Changes in reflex latencies and conduction velocities quickly resolve once the subject gets up from bed.

Exercise

The cycle ergometry test is performed to assess aerobic capacity. Maximum aerobic capacity decreased significantly after 30 days of bed rest. By the 12th day after bed rest, maximum aerobic capacity continued to be less than pre-bed rest, but was significantly improved from immediate post bed rest values. Isokinetic testing is used to evaluate changes in muscle strength and endurance following bed rest. Isokinetic muscle strength and muscle endurance are measured pre- and post-bed rest to quantify changes in the major locomotor and postural muscles. Testing evaluates extensor and flexor muscles of the lower extremity and trunk. In this study, knee flexor and extensor muscles showed the greatest changes. Strength values for these muscles demonstrated a decrease after 30 days of bed rest.

Nutrition

Nutritional status of the 30-day bed rest subjects was assessed through blood and urine biochemical testing profiles before and after bed rest. Assessment of bone analytes indicated that while bone formation tends to remain unchanged

	during bed rest, bone resorption significantly increases. Bone resorption showed an approximate 50%-75% increase between pre- and post- bed rest. Urinary sodium excretion patterns match dietary consistency, and fluid shifts upon entering and concluding bed rest. As fluid is lost early in bed rest, urine sodium concentrations show a relative increase. After bed rest as subjects retain more fluids due to re-adaptation to the vertical position, urine sodium concentration show a relative decrease. In bed rest studies, subjects with starting ferritin concentrations defined as low have been prescribed iron supplementation. These studies have shown small changes in iron status measurements, the most consistent being a drop in hematocrit and hemoglobin after subjects get up from bed, suggesting an impact of plasma volume replacement in the post bed rest phase. Similar findings have been seen after spaceflight.
Bibliography Type:	Description: (Last Updated: 05/16/2019)
Articles in Peer-reviewed Journals	Caiozzo VJ, Haddad F, Lee S, Baker M, Paloski W, Baldwin KM. "Artificial gravity as a countermeasure to microgravity: a pilot study examining the effects on knee extensor and plantar flexor muscle groups. " J Appl Physiol. 2009 Jul;107(1):39-46. Epub 2009 Mar 12. PMID: 19286573 ; http://dx.doi.org/10.1152/japplphysiol.91130.2008 , Jul-2009
Articles in Peer-reviewed Journals	Fong KJ, Arya M, Paloski WH. "Gender differences in cardiovascular tolerance to short radius centrifugation." J Gravit Physiol. 2007 Jul;14(1):P15-9. PMID: 18372686 , Jul-2007
Articles in Peer-reviewed Journals	Mehta SK, Crucian B, Pierson DL, Sams C, Stowe RP. "Monitoring immune system function and reactivation of latent viruses in the Artificial Gravity Pilot Study." J Gravit Physiol. 2007 Jul;14(1):P21-5. PMID: 18372687 , Jul-2007
Articles in Peer-reviewed Journals	Sakowski C, Starc V, Smith SM, Schlegel TT. "Sedentary long-duration head-down bed rest and ECG repolarization heterogeneity." Aviat Space Environ Med. 2011 Apr;82(4):416-23. PMID: 21485399 , Apr-2011
Articles in Peer-reviewed Journals	Seaton KA, Slack KJ, Sipes W, Bowie K. "Artificial gravity as a multi-system countermeasure: effects on cognitive function." J Gravit Physiol. 2007 Jul;14(1):P27-30. PMID: 18372688 , Jul-2007
Articles in Peer-reviewed Journals	Smith SM, Zwart SR, Heer MA, Baecker N, Evans HJ, Feiveson A, Shackelford LC, LeBlanc AD. "Effects of artificial gravity during bed rest on bone metabolism in humans." J Appl Physiol. 2009 Jul;107(1):47-53. Epub 2008 Dec 12. PMID: 19074572 , Jul-2009
Articles in Peer-reviewed Journals	Zwart SR, Crawford GE, Gillman PL, Kala G, Rodgers AS, Rogers A, Inniss AM, Rice BL, Ericson K, Coburn S, Bourbeau Y, Hudson E, Mathew G, Dekerlegand DE, Sams CF, Heer MA, Paloski WH, Smith SM. "Effects of 21 days of bed rest, with or without artificial gravity, on nutritional status of humans." J Appl Physiol. 2009 Jul;107(1):54-62. Epub 2008 Dec 12. PMID: 19074571 , Jul-2009
Articles in Peer-reviewed Journals	Zwart SR, Booth SL, Peterson JW, Wang Z, Smith SM. "Vitamin K status in spaceflight and ground-based models of spaceflight." J Bone Miner Res. 2011 May;26(5):948-54. http://dx.doi.org/10.1002/jbmr.289 ; PMID: 21541997 , May-2011